

Neural Network Preceptors: How Deep Learning can Help Solve AI Problems

This article, using an example, shows how a deep learning mechanism can help solve an artificial intelligence (AI) problem. It gives a good insight on how layers in training data can be formed.

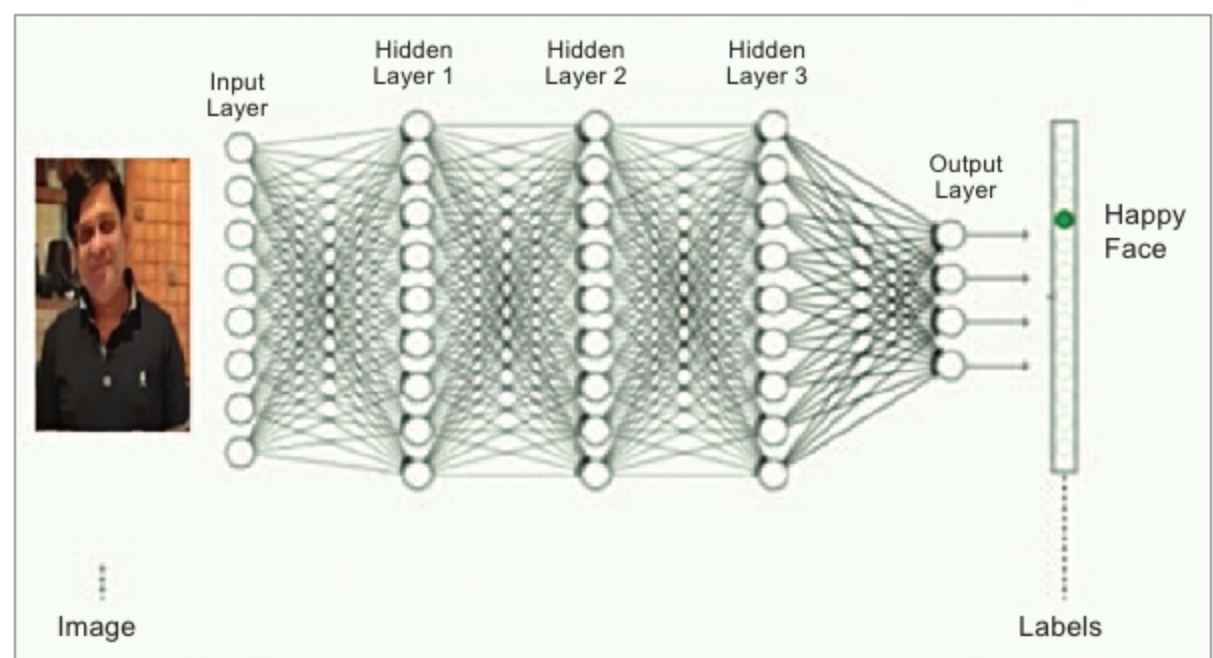
Creating an artificial intelligence (AI) system comes with many challenges and the biggest of all is to see how to get closer to human thinking and solve a problem the way a human will. Human or cognitive level of thinking is the most difficult, and to achieve that, it is necessary to make use of deep learning mechanism. Since convolution neural network (CNN) is the core of the deep learning mechanism, it allows adding desired intelligence to a system.

A neural network has layers of preceptors or logics/algorithms that can be written. Hence, the more layers of this logic one adds, the more the solution gets closer to cognitive thinking. The most important step here is back propagation, which helps analyse whatever has been done in all those layers. It then provides additional help by giving three times more accurate results.

Back propagation

Let us take the classic example of the lift versus staircase. Here, apart from all other aspects, it still depends on the mood of the person—whether the person will consider taking the lift or stairs—and this can change every day.

We create infinite scenarios, yet everyday being a new day we cannot be sure what will the mood be today, but we can still try to solve this and



How a deep learning mechanism reads a happy face

make it as accurate as we can using the following steps.

Human mood capture. How do we capture this mood and decide what is the person's mood today? We cannot read anyone's mind, but this can be achieved through face reading, which can give us facial expression and emotions, and body language, which can give us more information on the mood.

Face reading. Face reading or face detection software can help capture the emotions of the face, such as happy face, sad face or other. (There are many different types of emotions that can help in capturing a person's mood.) In this case, instead of photos, video clips of the person would be more helpful in making accurate predictions of the mood.

Note that, there are hundreds of different emotions and emojis, and if we consider all of them, it will not solve the issue. It would make up huge amounts of similar data, since there are many emotions that are related and similar to each other. Only a few basic emotions need to be checked, which are obvious in nature. It works in the same way as it does with colours, where there are infinite colours, but we take three primary colours and then differentiate them accordingly. Hence, the seven basic emotions to be considered are joy, sadness, anger fear, disgust, neutral and surprise.

According to psychology, facial expressions are checked with eyes, nose and mouth changes. Therefore the facial

